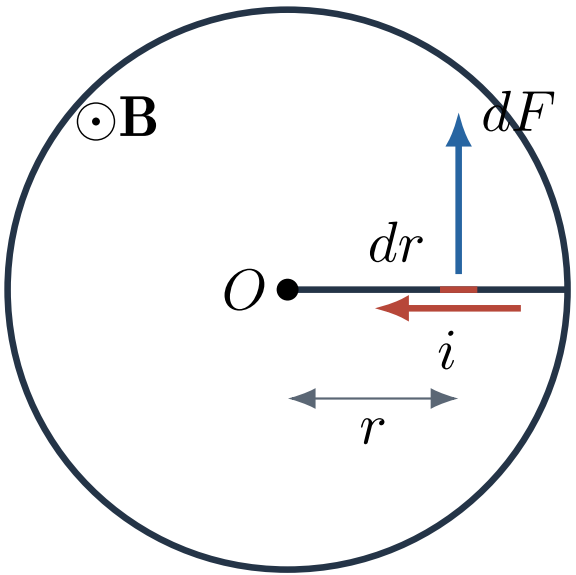
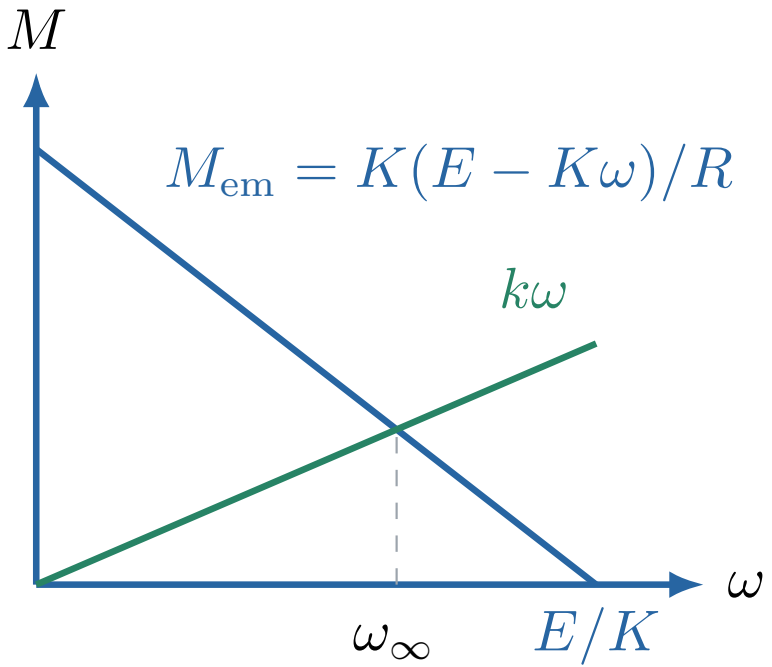


$$dM = r\, iB\, dr$$



$$K = Br_0^2/2$$